IMPLEMENTACIÓN DE VPN IPSEC SITE-TO-SITE

Actividad 6

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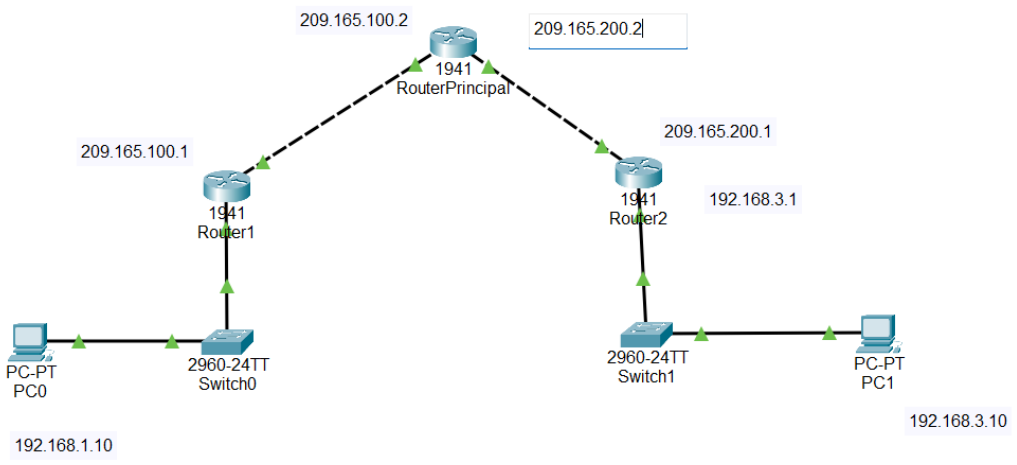
Objetivo

Implementar una **VPN Site-to-Site utilizando el protocolo IPsec**, con el fin de establecer una comunicación segura entre dos redes remotas a través de una red pública. La práctica permite comprender el funcionamiento de políticas ISAKMP, transform-set, listas de control de acceso (ACL) y mapas criptográficos, asegurando la confidencialidad, integridad y autenticación de los datos transmitidos.

Topologia

- Tres routers 1941 (ISP, R1, R2)
- Dos switch 2960 (S1, S2)
- Dos PC´s



Desarrollo

Configuración Inicial

El primer paso es realizar las conexiones entre los dispositivos mediante los cables correspondientes, cable cruzado para dispositivos diferentes y cable directo para dispositovos del mismo tipo.

Router principal:

Acceso al router
Se abrió el dispositivo router en Cisco Packet Tracer y se ingresó a la pestaña CLI para comenzar la configuración mediante línea de comandos.

Ingreso al modo privilegiado
Para poder realizar cambios en la configuración del dispositivo, se ingresó al modo de configuración global

Se configuró el nombre del Reuter como RISP para facilitar su identificación dentro de la topología de red

Configuración de la interfaz GigabitEthernet 0/0
Se accedió a la interfaz GigabitEthernet0/0 y se le asignó la dirección IP correspondiente, además de habilitarla

Configuración de la interfaz GigabitEthernet 0/1
Posteriormente, se configuró la interfaz GigabitEthernet0/1 con su respectiva dirección IP y se activó

Configuración de la ruta estática por defecto
Finalmente, se configuró una ruta estática por defecto para enviar el tráfico destinado a redes desconocidas

```
R1#
R1#enable
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#hostname RIPS
RIPS(config)#interface G0/0
RIPS(config-if)#ip address 209.165.100.2 255.255.255.0
RIPS(config-if)#no shut
RIPS(config-if)#interface G0/1
RIPS(config-if)#ip address 209.165.200.2 255.255.255.0
RIPS(config-if)#no shut
RIPS(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.100.2
%Invalid next hop address (it's this router)
RIPS(config)#
```

Ahora configuraremos el Router 1 (R1):

Ingreso al modo privilegiado

Acceso al modo de configuración global
Asignación del nombre del router

Configuración de la interfaz GigabitEthernet 0/1

Configuración de la interfaz GigabitEthernet 0/0

Configuración de la ruta estática por defecto

```
router#
router#enable
router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
router(config)#hostname R1
R1(config)#interface G0/1
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#interface G0/0
R1(config-if)#ip address 209.165.100.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.100.2
R1(config)#
```

De igual forma configuramos el router 2 con sus respectivas ip

```
R2#
R2#enable
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname R2
R2(config)#interface G0/1
R2(config-if)#ip address 192.168.3.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#interface G0/0
R2(config-if)#ip address 209.165.200.1 255.255.255.0
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#ip address 209.165.200.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.100.2
R2(config)#
```

2. Licencia de seguridad

En cada router instalamos la licencia de seguridad:

RISP:

```
RISP(config)#license boot module c1900 technology-package securityK9
PLEASE  READ THE  FOLLOWING TERMS  CAREFULLY. INSTALLING THE LICENSE OR
LICENSE  KEY  PROVIDED FOR  ANY CISCO  PRODUCT  FEATURE  OR  USING SUCH
PRODUCT  FEATURE  CONSTITUTES  YOUR  FULL ACCEPTANCE  OF  THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO  BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.

Use of this product feature requires  an additional license from Cisco,
together with an additional  payment.  You may use this product feature
on an evaluation basis, without payment to Cisco, for 60 days. Your use
of the  product,  including  during the 60 day  evaluation  period,  is
subject to the Cisco end user license agreement
http://www.cisco.com/en/US/docs/general/warranty/English/EU1KEN_.html
If you use the product feature beyond the 60 day evaluation period, you
must submit the appropriate payment to Cisco for the license. After the
60 day  evaluation  period,  your  use of the  product  feature will be
governed  solely by the Cisco  end user license agreement (link above),
together  with any supplements  relating to such product  feature.  The
above  applies  even if the evaluation  license  is  not  automatically
terminated  and you do  not receive any notice of the expiration of the
evaluation  period.  It is your  responsibility  to  determine when the
evaluation  period is complete and you are required to make  payment to
Cisco for your use of the product feature beyond the evaluation period.

Your  acceptance  of  this agreement  for the software  features on one
product  shall be deemed  your  acceptance  with  respect  to all  such
software  on all Cisco  products  you purchase  which includes the same
software.  (The foregoing  notwithstanding, you must purchase a license
for each software  feature you use past the 60 days evaluation  period,
so  that  if you enable a software  feature on  1000  devices, you must
purchase 1000 licenses for use past  the 60 day evaluation period.)

ACCEPT? [yes/no]: yes
% use 'write' command to make license boot config take effect on next boot

RISP(config)#: %IOS_LICENSE_IMAGE_APPLICATION-6-LICENSE_LEVEL: Module name = C1900 Next reboot
level = securityk9 and License = securityk9

RISP(config)#
```

R1:

```
R1>enable
R1#
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#license boot module c1900 technology-package securityK9
PLEASE  READ THE  FOLLOWING TERMS  CAREFULLY. INSTALLING THE LICENSE OR
LICENSE  KEY  PROVIDED FOR  ANY CISCO  PRODUCT  FEATURE  OR  USING SUCH
PRODUCT  FEATURE  CONSTITUTES  YOUR  FULL ACCEPTANCE  OF  THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO  BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.
```

R2:

```
R2(config)#license boot module c1900 technology-package securityK9
PLEASE  READ THE  FOLLOWING TERMS  CAREFULLY. INSTALLING THE LICENSE OR
LICENSE  KEY  PROVIDED FOR  ANY CISCO  PRODUCT  FEATURE  OR  USING SUCH
PRODUCT  FEATURE  CONSTITUTES  YOUR  FULL ACCEPTANCE  OF  THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO  BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.

Use of this product feature requires  an additional license from Cisco,
together with an additional  payment.  You may use this product feature
```

Implementación de ACL´s

Listas de Control de Acceso (ACL)

Se configura el router R1:

```
R1(config)#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

R2:

```
R2(config)#access-list 100 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
```

Phase 01: ISAKMP policy

Ahora ciframos ambos el encabezado de router r1 y r2 utilizando la politica ISAKMP

```
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#autentication pre-share
                        ^
% Invalid input detected at '^' marker.

R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#exit
R1(config)#crypto isakmp key key1 address 209.165.200.1
R1(config)#
```

```
R2(config-isakmp)#encryption aes 256 authentication pre-share
                                     ^
% Invalid input detected at '^' marker.

R2(config-isakmp)#encryption aes 256 authentication pre-share
                                     ^
% Invalid input detected at '^' marker.

R2(config-isakmp)#encryption aes 256
R2(config-isakmp)#authentication pre-share
R2(config-isakmp)#group 5
R2(config-isakmp)#exit
R2(config)#crypto isakmp key key1 address 209.165.200.1
R2(config)#
```

La Phase 1 establece un canal seguro entre los routers, autenticando a los dispositivos y negociando los parámetros de seguridad antes de intercambiar datos.

Phase 2: IPSec transform-set

Configuracion de Cifrado del contenido del paquete para su traslado en ambos routers R1-R2

```
R1(config)#crypto ipsec transform-set R1->R2 esp-aes 256 esp-sha-hmac
```

```
R2(config)#crypto ipsec transform-set R2->R1 esp-aes 256 esp-sha-hmac
R2(config)#
```

La Phase 2 se encarga de cifrar el contenido de los paquetes que viajan por la VPN, garantizando la confidencialidad e integridad de la información.

Creación de mapa criptográfico:

```
R1(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)#set peer 209.165.200.1
R1(config-crypto-map)#set pfs group5
R1(config-crypto-map)#
```

```
R2(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R2(config-crypto-map)#set peer 209.165.100.1
R2(config-crypto-map)#set pfs group5
R2(config-crypto-map)#set security-association lifetime seconds 86400
R2(config-crypto-map)#set transform-set R2->R1
R2(config-crypto-map)#match address 100
R2(config-crypto-map)#
```

El mapa criptográfico enlaza todos los elementos de la VPN, indicando qué tráfico se cifrará, cómo se cifrará y hacia qué router remoto se enviará

Aplicacion de mapa criptografico:

Al aplicar el mapa criptográfico a la interfaz, se activa la VPN, permitiendo que el tráfico definido en la ACL sea protegido mediante IPsec.

```
R1(config)#
R1(config)#
R1(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)#set peer 209.165.200.1
R1(config-crypto-map)#set pfs group5
R1(config-crypto-map)#exit
R1(config)#int G0/0
R1(config-if)#crypto map IPSEC-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)#
```

```
R2(config-crypto-map)#
R2(config-crypto-map)#exit
R2(config)#int G0/0
R2(config-if)#crypto map IPSEC-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R2(config-if)#
```

El ping exitoso confirma que la VPN Site-to-Site funciona correctamente y que la comunicación entre redes remotas se realiza de manera segura